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Standards & Labels

BEE Implementation

The presence of a large number of unorganised suppliers with low-priced inverter offerings compelled the organised players to demand for standards and labels for inverters, especially at the low end. To put an end to this unhealthy competition, the Bureau of Energy Efficiency (BEE), in association with Collaborative Labeling and Appliance Standards Program and PricewaterhouseCoopers (CLASP/PwC), initiated preparatory studies for standards and the labelling of domestic inverters up to 2 kVA.

Now the home UPS manufacturers need to improve the designs conforming to BEE specifications for 'star rating'. Though the specifications made by BEE

are very difficult to adhere to, with the arrival of new technologies, efficiencies should improve. Home UPS should have better charging algorithms so that batteries last longer and the environment is protected from the use of more lead.

With the implementation of BEE standards, consumers will benefit the most as they would get a higher power output from the inverter, and a longer power backup period due to the high ef-

iciency of the inverter, at rated full load.

BEE has not only specified the idle power consumption during mains bypass but also the charging-mode operation of the batteries. Considering that these factors will improve the input power factor and the efficiency of the system, consumers will benefit as they will need to charge their batteries less often, saving power significantly. This saved energy would mean financial gains as well.

What is UPS/inverter efficiency?

The energy efficiency of UPS system/inverter is the difference between the amount of energy that is used by a UPS system and the amount of useful energy that it delivers to power loads.

In all UPS systems, some amount of energy is lost as heat when it passes through the internal components of the UPS (transformers, rectifiers, inverters, etc). The amount of energy lost between the input and output can be significant when you consider the costs of the wasted energy. Energy efficiency advantages of as little as 1 per cent between one UPS and another can translate into thousands of rupees saved per year, depending on the size of the UPS.

New Markets

Beyond computer load and metros

Laptop and notebook sales have grown at a healthy rate of 65 per cent, while that of desktop PCs is just 4 per cent. This is definitely going to have a negative impact on UPS sales, but this holds true only for internal-battery UPS.

External-battery UPS used for extended backups in various applications will continue to grow. There are various factors instrumental in fuelling the UPS growth. The server market is growing at

a brisk rate of 22 per cent. Industrial automation and process industry is witnessing a phenomenal growth and the small and medium businesses (SMB) sector is actively modernising. With more process automation using SCADA/DCS and many other latest systems, UPS is looked beyond computer load in the present context.

In fact, a Frost & Sullivan report expects the Indian UPS market to grow at the rate of 13.4 per cent annually to reach the market size of \$1316.5 million by 2014. This growth will be driven by

demands from major sectors such as banking, financial services and insurance, telecom, retail, IT/ITES, manufacturing and infrastructure.

Manufacturers have set their eyes on B and C class cities to make up for the reduction in demand in metro cities. With the increase in PC use in these cities, the sales of inverter and UPS has increased. The mushrooming up of manufacturing and service units, malls, food joints and even hotels in small cities is fuelling the demand.